



# Organisms and Populations

## Unit X ECOLOGY - Unit Introduction

Diversity is not only a characteristic of living organisms but also of content in biology textbooks. Biology is presented either as **botany, zoology and microbiology** or as **classical and modern**; the latter is a euphemism for molecular aspects of biology. Luckily we have many threads which weave the different areas of biological information into a unifying principle, and **ecology** is one such thread - it gives us a holistic perspective to biology. The essence of biological understanding is to know how organisms, while remaining an individual, interact with other organisms and physical habitats as a group and hence behave like organised wholes, i.e., **population, community, ecosystem** or even as the whole **biosphere**. Ecology explains to us all this. A particular aspect of this is the study of anthropogenic environmental degradation and the socio-political issues it has raised. This unit describes as well as takes a critical view of the above aspects.

- Unit X (ten) - **ECOLOGY** - contains three chapters: Chapter 11 *Organisms and Populations*; Chapter 12 *Ecosystem*; Chapter 13 *Biodiversity and Conservation*.

## Profile RAMDEO MISRA (1908-1998)

- **Ramdeo Misra is revered as the Father of Ecology in India.** Born on **26 August 1908**; dates **1908-1998**.
- Obtained **Ph.D in Ecology (1937)** under Prof. **W. H. Pearsall, FRS**, from **Leeds University** in UK.
- Established teaching and research in ecology at the **Department of Botany of the Banaras Hindu University, Varanasi**.
- His research laid the foundations for understanding of **tropical communities and their succession, environmental responses of plant populations** and **productivity and nutrient cycling in tropical forest and grassland ecosystems**.
- Formulated the **first postgraduate course in ecology in India**.
- **Over 50 scholars** obtained Ph. D degree under his supervision and moved on to other universities and research institutes to initiate ecology teaching and research across the country.
- Honoured with the Fellowships of the **Indian National Science Academy** and **World Academy of Arts and Science**, and the prestigious **Sanjay Gandhi Award in Environment and Ecology**.
- Due to his efforts, the Government of India established the **National Committee for Environmental Planning and Coordination (1972)**, which in later years paved the way for the establishment of the **Ministry of Environment and Forests (1984)**.

## Ch 11 ORGANISMS AND POPULATIONS - Chapter Opener

Biological organisation spans macromolecules, cells, tissues, organs, organisms, populations, communities, ecosystems and biomes; ecology examines interactions at the organism, population, community and biome levels. At any level of biological organisation we can ask **two types of questions**. When we hear the bulbul singing early morning in the garden, we may ask '*How does the bird sing?*' or '*Why does the bird sing?*'

| Question type | What it seeks                           | Answer for the singing bulbul                                                                                                                  |
|---------------|-----------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------|
| 'How-type'    | The <b>mechanism</b> behind the process | For the first question, the answer <b>might</b> be in terms of the operation of the <b>voice box</b> and the <b>vibrating bone</b> in the bird |
| 'Why-type'    | The <b>significance</b> of the process  | For the second question, the answer <b>may lie</b> in the bird's <b>need to communicate with its mate during breeding season</b>               |

① **[NOTE]** NCERT prints "*chick spurs*" in this list of questions; the wording is reproduced as printed. Note the qualifier in the first question: *night-blooming flowers are **generally** white, not always.*

Ecology is the study of interactions between organisms and the abiotic (physico-chemical) and biotic (other species) components of their environment.

- The four levels are organisms, populations, communities and biomes; this chapter focuses on organisms and populations.

In nature, we **rarely** find isolated, single individuals of any species; **majority** of them live in groups in a well defined geographical area, share or compete for similar resources, potentially interbreed and thus constitute a **population**.

- Although the term **interbreeding** implies sexual reproduction, a group of individuals resulting from even **asexual reproduction** is also **generally** considered a population for the purpose of ecological studies.
- Examples of a population: **all the cormorants in a wetland, rats in an abandoned dwelling, teakwood trees in a forest tract, bacteria in a culture plate and lotus plants in a pond.**
- Although an **individual organism** is the one that has to cope with a changed environment, it is at the **population level** that **natural selection** operates to evolve the desired traits. Population ecology is, therefore, an important area because it links **ecology to population genetics and evolution**.

A population has certain attributes whereas an individual organism does not. An individual may have **births and deaths**, but a population has **birth rates and death rates**. In a population these rates refer to **per capita** births and deaths, i.e. the rates expressed are change in numbers (increase or decrease) **with respect to members of the population**.

| Attribute                                                                                                                                   | An individual                | A population                                                                                                              |
|---------------------------------------------------------------------------------------------------------------------------------------------|------------------------------|---------------------------------------------------------------------------------------------------------------------------|
| Births / deaths                                                                                                                             | Has births and deaths        | Has <b>birth rates</b> and <b>death rates</b> (per capita)                                                                |
| Sex                                                                                                                                         | Is either a male or a female | Has a <b>sex ratio</b> (e.g., 60 per cent of the population are females and 40 per cent males)                            |
| Age                                                                                                                                         | Has one age at a time        | Is composed of individuals of <b>different ages</b> ; the <b>age distribution</b> can be plotted as an <b>age pyramid</b> |
| Worked example                                                                                                                              |                              | Calculation                                                                                                               |
| A pond had <b>20</b> lotus plants last year; through reproduction <b>8</b> new plants are added, taking the current population to <b>28</b> |                              | 8/20                                                                                                                      |
| <b>4</b> individuals in a laboratory population of <b>40</b> fruitflies died during a specified time interval, say <b>a week</b>            |                              | 4/40                                                                                                                      |
|                                                                                                                                             |                              | <b>Rate</b>                                                                                                               |
|                                                                                                                                             |                              | <b>0.4 offspring per lotus per year</b> (birth rate)                                                                      |
|                                                                                                                                             |                              | <b>0.1 individuals per fruitfly per week</b> (death rate)                                                                 |

A population at any given time is composed of individuals of different ages. If the **age distribution** (per cent individuals of a given age or age group) is plotted for the population, the resulting structure is called an **age pyramid** (Figure 11.1). For human population, the age pyramids **generally** show age distribution of **males and females** in a diagram. The three age classes plotted are **Pre-reproductive, Reproductive** and **Post-reproductive**.

| Pyramid shape (as labelled in Figure 11.1)                                                                                                                                                                                                                                                                                        | Growth status of the population                               |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------|
| <b>Expanding</b> - broad Pre-reproductive base                                                                                                                                                                                                                                                                                    | (a) whether it is <b>growing</b>                              |
| <b>Stable</b> - Pre-reproductive and Reproductive classes about equal                                                                                                                                                                                                                                                             | (b) <b>stable</b> (the Summary calls this <b>stationary</b> ) |
| <b>Declining</b> - narrow Pre-reproductive base                                                                                                                                                                                                                                                                                   | (c) <b>declining</b>                                          |
| ① <b>[NOTE]</b> The <b>shape</b> of the pyramids reflects the growth status of the population - (a) whether it is growing, (b) stable or (c) declining. The chapter Summary uses the word <b>stationary</b> where the body and the figure label say <b>Stable</b> ; both wordings are NCERT's, so recognise either in a question. |                                                               |

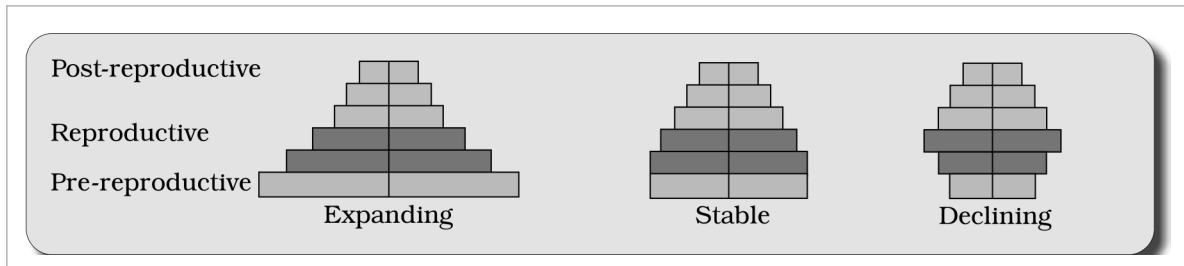


Fig. 11.1 - Representation of age pyramids for human population. Each pyramid is read as three stacked age classes - Pre-reproductive (bottom), Reproductive (middle) and Post-reproductive (top) - with males and females on the two sides; the three shapes are labelled Expanding, Stable and Declining.

#### 11.1.1 Population size and population density (N)

The **size** of the population tells us a lot about its status in the habitat. Whatever ecological processes we wish to investigate in a population - be it the outcome of **competition** with another species, the impact of a **predator** or the effect of a **pesticide application** - we **always** evaluate them in terms of any change in the population size.

- The size, in nature, could be as low as **<10** (*Siberian cranes* at **Bharatpur wetlands** in any year) or go into **millions** (*Chlamydomonas* in a pond).

● **Population size**, technically called **population density** (designated as **N**), need **not necessarily** be measured in numbers only.

Although **total number** is **generally** the most appropriate measure of population density, it is **in some cases** either meaningless or difficult to determine:

| Problem with total number                                                                                                   | NCERT's case                                                                                                                                                                                                                                                         | Better measure                                                                      |
|-----------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------|
| Total number is <b>meaningless</b> - it hides the ecological role of a huge individual                                      | An area with <b>200 carrot grass</b> ( <i>Parthenium hysterophorus</i> ) plants but only a <b>single huge banyan tree with a large canopy</b> : calling banyan density low relative to carrot grass underestimates the enormous role of the Banyan in that community | <b>Per cent cover</b> or <b>biomass</b>                                             |
| Total number is <b>not easily adoptable</b> - the population is huge and counting is impossible or very time-consuming      | A <b>dense laboratory culture of bacteria in a petri dish</b> (embedded question: what is the best measure to report its density?)                                                                                                                                   | A measure that does not need head-counting, e.g. biomass or per cent cover          |
| <b>Sometimes</b> , for certain ecological investigations, there is no need to know the <b>absolute</b> population densities | The number of <b>fish caught per trap</b> is good enough measure of its total population density in the lake                                                                                                                                                         | <b>Relative densities</b> serve the purpose equally well                            |
| We are <b>mostly</b> obliged to estimate population sizes <b>indirectly</b> , without actually counting them or seeing them | The <b>tiger census</b> in our national parks and tiger reserves                                                                                                                                                                                                     | <b>Indirect evidence</b> - often based on <b>pug marks</b> and <b>fecal pellets</b> |

① **[NOTE]** Ecological effects of any factor on a population are **generally** reflected in its size (population density), which may be expressed in different ways - **numbers**, **biomass**, **per cent cover**, etc. - depending on the species.

#### 11.1.2 Population Growth

The size of a population for any species is **not a static parameter**. It keeps changing with time, depending on various factors including **food availability**, **predation pressure** and **adverse weather**. In fact, it is these changes in population density that give us some idea of what is happening to the population - whether it is **flourishing** or **declining**.

The density of a population in a given habitat during a given period fluctuates due to changes in **four basic processes**, two of which (**natality** and **immigration**) contribute to an **increase** in population density and two (**mortality** and **emigration**) to a **decrease**:

| Process (as labelled in Figure 11.2) | Symbol | NCERT definition                                                                                                                                      | Effect on Population Density (N) |
|--------------------------------------|--------|-------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------|
| (i) <b>Natality (B)</b>              | B      | The number of <b>births</b> during a given period in the population that are <b>added to the initial density</b>                                      | <b>Increase</b>                  |
| (iii) <b>Immigration (I)</b>         | I      | The number of individuals of the <b>same species</b> that have <b>come into</b> the habitat from elsewhere during the time period under consideration | <b>Increase</b>                  |
| (ii) <b>Mortality (D)</b>            | D      | The number of <b>deaths</b> in the population during a given period                                                                                   | <b>Decrease</b>                  |
| (iv) <b>Emigration (E)</b>           | E      | The number of individuals of the population who <b>left the habitat and gone elsewhere</b> during the time period under consideration                 | <b>Decrease</b>                  |

So, if **N** is the population density at time **t**, then its density at time **t + 1** is

$$N_{t+1} = N_t + [(B + I) - (D + E)]$$

You can see from the above equation (Fig. 11.2) that **population density will increase** if the number of births plus the number of immigrants (**B + I**) is **more than** the number of deaths plus the number of emigrants (**D + E**).

- Under **normal conditions**, **births and deaths** are the most important factors influencing population density, the other two factors assuming importance **only under special conditions**.
- For instance, if a **new habitat is just being colonised**, **immigration** may contribute more significantly to population growth than birth rates.

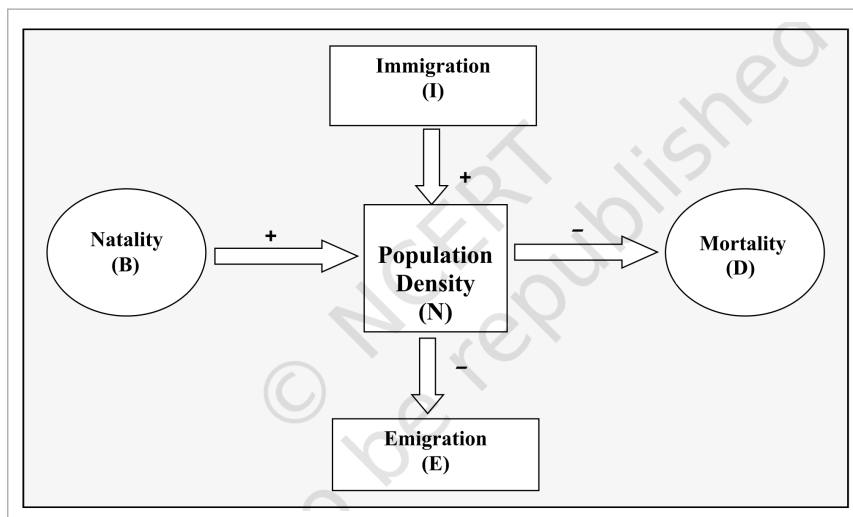


Fig. 11.2 - The four processes acting on Population Density (N): Natality (B) and Immigration (I) add to it, Mortality (D) and Emigration (E) subtract from it. NCERT gives this figure no caption text beyond the number; the '+' and '-' signs on its arrows are the same sign convention used in the equation above.

☆ **[MEMORY AID - not in NCERT]** **IN** raises density, **ME** lowers it: **Immigration + Natality in; Mortality + Emigration out.**

#### 11.1.2 Growth Models

Population growth follows predictable exponential or logistic patterns, while natural populations face restraints on growth.

##### (i) Exponential growth

**Resource (food and space) availability** is obviously essential for the **unimpeded** growth of a population. **Ideally**, when resources in the habitat are **unlimited**, each species has the ability to realise **fully** its **innate potential** to grow in number, as **Darwin** observed while developing his theory of **natural selection**. Then the population grows in an **exponential or geometric** fashion.

① **[NOTE]** The chapter Summary hedges this statement: "When resources are unlimited, the growth is **usually** exponential but when resources become progressively limiting, the growth pattern turns logistic." The body states the exponential case without the hedge - keep NCERT's **usually** when the Summary wording is quoted.

If in a population of size **N** the birth rates (**not total number but per capita births**) are represented as **b** and death rates (again, **per capita** death rates) as **d**, then the increase or decrease in **N** during a unit time period **t** (**dN/dt**) will be:

$$dN/dt = (b - d) \times N$$

Let **(b - d) = r**, then

$$dN/dt = rN$$

- The **r** in this equation is called the '**intrinsic rate of natural increase**' and is a very important parameter chosen for **assessing impacts of any biotic or abiotic factor** on population growth. The Summary adds that **r** is a measure of the **inherent potential of a population to grow**.

| Population                                                                                                                                           | r value |
|------------------------------------------------------------------------------------------------------------------------------------------------------|---------|
| Norway rat                                                                                                                                           | 0.015   |
| Flour beetle                                                                                                                                         | 0.12    |
| Human population in India, 1981                                                                                                                      | 0.0205  |
| ① [NOTE] NCERT activity: <b>find out what the current r value is</b> . For calculating it, you need to know the <b>birth rates and death rates</b> . |         |

The above equation describes the **exponential or geometric growth pattern** of a population (Figure 11.3) and results in a **J-shaped curve** when we plot **N in relation to time**. If you are familiar with basic calculus, you can derive the **integral form** of the exponential growth equation as:

$$N_t = N_0 e^{rt}$$

| Term     | Meaning                                         |
|----------|-------------------------------------------------|
| $N_t$    | Population density <b>after time t</b>          |
| $N_0$    | Population density <b>at time zero</b>          |
| <b>r</b> | <b>Intrinsic rate of natural increase</b>       |
| <b>e</b> | The base of <b>natural logarithms (2.71828)</b> |

**Any species growing exponentially under unlimited resource conditions can reach enormous population densities in a short time.** Darwin showed how even a **slow growing animal like elephant** could reach enormous numbers **in the absence of checks**. The following anecdote is popularly narrated to demonstrate dramatically how fast a huge population could build up when growing exponentially.

- 1 The **king and the minister** sat for a **chess game**. The king, confident of winning, was ready to accept **any bet** proposed by the minister.
- 2 The minister humbly asked only for **some wheat grains**, the quantity to be calculated by placing on the chess board **one grain in Square 1, two in Square 2, four in Square 3, eight in Square 4**, and so on - **doubling each time** the previous quantity of wheat on the next square **until all the 64 squares were filled**.
- 3 The king accepted the seemingly silly bet and started the game, but **unluckily for him, the minister won**.
- 4 By the time he covered **half the chess board**, the king realised to his dismay that **all the wheat produced in his entire kingdom pooled together** would still be **inadequate** to cover all the **64 squares**.

If one Paramecium doubled by binary fission each day, it would reach an enormous population in 64 days, provided food and space remained unlimited.

## (ii) Logistic growth

**No population of any species in nature** has at its disposal **unlimited** resources to permit exponential growth. This leads to **competition between individuals for limited resources**. Eventually, the '**fittest**' individual **will survive and reproduce**. The governments of many countries have also realised this fact and introduced various **restraints** with a view to limit human population growth.

- In nature, a given habitat has enough resources to support a **maximum possible number**, beyond which **no further growth is possible**. This limit is nature's **carrying capacity (K)** for that species in that habitat.

A population growing in a habitat with **limited resources** shows **initially a lag phase, followed** by phases of **acceleration** and **deceleration** and **finally an asymptote**, when the population density reaches the carrying capacity:

- 1 **Lag phase** - the initial phase.
- 2 **Acceleration** - growth speeds up.
- 3 **Deceleration** - growth slows down.
- 4 **Asymptote** - reached when the **population density reaches the carrying capacity (K)**.

A plot of **N in relation to time (t)** results in a **sigmoid curve**. This type of population growth is called **Verhulst-Pearl Logistic Growth** (Figure 11.3) and is described by the following equation:

$$dN/dt = rN [(K - N)/K]$$

| Term | Meaning                            |
|------|------------------------------------|
| N    | Population density at time t       |
| r    | Intrinsic rate of natural increase |
| K    | Carrying capacity                  |

- Since resources for growth for **most** animal populations are **finite** and become limiting **sooner or later**, the **logistic growth model is considered a more realistic one**.

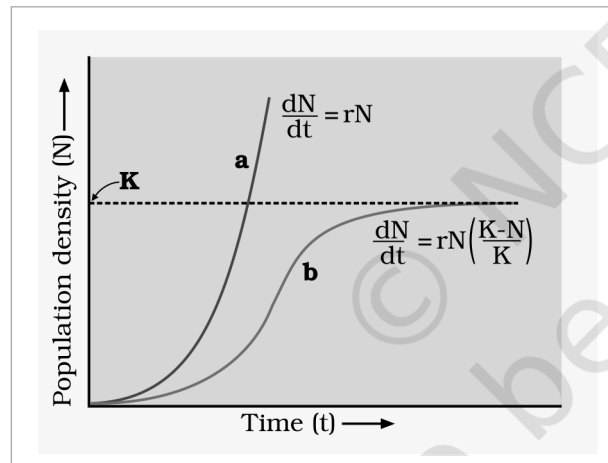


Fig. 11.3 - Population growth curve: **a** when responses are not limiting the growth, plot is exponential; **b** when responses are limiting the growth, plot is logistic; **K** is carrying capacity. Axes: Population density (N) against Time (t); curve **a** carries  $dN/dt = rN$  and curve **b** carries  $dN/dt = rN (K-N)/K$ .

- ① **[NOTE]** Both models are limited in the end: as the Summary puts it, **in either case, growth is ultimately limited by the carrying capacity of the environment**. The two curves in Figure 11.3 differ only in whether resources are limiting - exponential (**a**, J-shaped) versus logistic (**b**, sigmoid, flattening at **K**).

| Feature           | Exponential (geometric) growth                            | Logistic (Verhulst-Pearl) growth                                                                          |
|-------------------|-----------------------------------------------------------|-----------------------------------------------------------------------------------------------------------|
| Resources         | <b>Unlimited</b>                                          | <b>Limited</b> - competition between individuals                                                          |
| Equation          | $dN/dt = rN$ ; integral form $N_t = N_0 e^{rt}$           | $dN/dt = rN [(K - N)/K]$                                                                                  |
| Curve             | <b>J-shaped</b> (curve <b>a</b> of Figure 11.3)           | <b>Sigmoid</b> (curve <b>b</b> of Figure 11.3), with lag, acceleration, deceleration and asymptote phases |
| Carrying capacity | Not reached - density can become enormous in a short time | Growth stops at <b>K</b>                                                                                  |
| Realism           | Ideal case only                                           | <b>More realistic</b> for most animal populations                                                         |

- ① **[NOTE]** NCERT activity: gather from **Government Census data** the population figures for **India for the last 100 years**, plot them and check **which growth pattern is evident**.

Populations evolve to **maximise their reproductive fitness**, also called **Darwinian fitness (high r value)**, in the habitat in which they live. Under a particular set of **selection pressures**, organisms evolve towards the **most efficient reproductive strategy**.

| Life history trait           | One extreme                                                                     | The other extreme                                                             |
|------------------------------|---------------------------------------------------------------------------------|-------------------------------------------------------------------------------|
| Number of breeding events    | Breed <b>only once</b> in their lifetime - <b>Pacific salmon fish, bamboo</b>   | Breed <b>many times</b> during their lifetime - <b>most birds and mammals</b> |
| Number and size of offspring | A <b>large number of small-sized</b> offspring - <b>Oysters, pelagic fishes</b> | A <b>small number of large-sized</b> offspring - <b>birds, mammals</b>        |

Life-history traits evolve under the abiotic and biotic constraints of a habitat, a subject of continuing ecological study.

No natural habitat contains just one species; every species depends on at least one other species for food.

- Even autotrophs depend on other organisms: soil microbes return nutrients, and animal pollinators enable pollination.
- In nature, animals, plants and microbes **do not and cannot live in isolation** but interact in various ways to form a **biological community**. Even in **minimal communities**, **many** interactive linkages exist, although **all may not** be readily apparent.

**Interspecific interactions** arise from the interaction of **populations of two different species**. They could be **beneficial, detrimental or neutral** (neither harm nor benefit) **to one of the species or both**. Assigning a '+' sign for beneficial interaction, '-' sign for detrimental and 0 for neutral interaction, we get all the possible outcomes of interspecific interactions (Table 11.1).

Table 11.1

Population Interactions

| Species A | Species B | Name of Interaction |
|-----------|-----------|---------------------|
| +         | +         | <b>Mutualism</b>    |
| -         | -         | <b>Competition</b>  |
| +         | -         | <b>Predation</b>    |
| +         | -         | <b>Parasitism</b>   |
| +         | 0         | <b>Commensalism</b> |
| -         | 0         | <b>Amensalism</b>   |

- **Both** the species **benefit** in **mutualism** and **both lose** in **competition** in their interactions with each other.
- In **both parasitism and predation only one** species benefits (**parasite** and **predator**, respectively) and the interaction is **detrimental to the other species** (**host** and **prey**, respectively).
- The interaction where **one species is benefitted and the other is neither benefitted nor harmed** is called **commensalism**.
- In **amensalism**, on the other hand, **one species is harmed whereas the other is unaffected**.
- **Predation, parasitism and commensalism** share a common characteristic - **the interacting species live closely together**.

☆ **[MEMORY AID - not in NCERT]** Read Table 11.1 by signs, not by names: ++ mutualism, -- competition, +- predation and parasitism, +0 commensalism, -0 amensalism.

Predation transfers energy fixed by plants to higher trophic levels.

- Predation is not limited to tiger and deer: a sparrow eating seeds is also a predator.
- Although animals eating plants are categorised separately as **herbivores**, they are, **in a broad ecological context, not very different from predators**.

Besides acting as '**conduits**' for **energy transfer across trophic levels**, predators play other important roles:

- **They keep prey populations under control**. But for predators, prey species could achieve **very high population densities** and cause **ecosystem instability**. (The Summary hedges this as **some** predators help in controlling their prey populations.)
  - When certain **exotic species** are introduced into a geographical area, they become **invasive** and start spreading fast because the **invaded land does not have its natural predators**.
  - The **prickly pear cactus** introduced into **Australia in the early 1920's** caused havoc by spreading rapidly into **millions of hectares of rangeland**. The invasive cactus was brought under control **only** after a **cactus-feeding predator (a moth)** from its natural habitat was introduced into the country.
  - **Biological control methods** adopted in **agricultural pest control** are based on the ability of the **predator to regulate prey population**.
- **Predators also help in maintaining species diversity** in a community, by **reducing the intensity of competition** among competing prey species.
  - In the **rocky intertidal communities of the American Pacific Coast** the starfish *Pisaster* is an important predator. In a field experiment, when **all the starfish were removed** from an enclosed intertidal area, **more than 10 species of invertebrates became extinct within a year**, because of **inter-specific competition**.

① **[NOTE]** If a predator is **too efficient** and **overexploits its prey**, then the prey might become extinct and, following it, the predator will **also** become extinct for lack of food. **This is the reason why predators in nature are 'prudent'**.

Prey species have evolved various defenses to lessen the impact of predation:

| Prey defence                                                                                                                                            | NCERT example                                                                     |
|---------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------|
| <b>Cryptic colouration (camouflage)</b> - to avoid being detected easily by the predator                                                                | <b>Some species of insects and frogs</b>                                          |
| <b>Being poisonous</b> - and therefore avoided by the predators                                                                                         | <b>Some prey species</b>                                                          |
| <b>Distastefulness</b> from a <b>special chemical</b> present in the body - acquired during the <b>caterpillar stage by feeding on a poisonous weed</b> | The <b>Monarch butterfly</b> , highly distasteful to its predator ( <b>bird</b> ) |

**For plants, herbivores are the predators**. Nearly **25 per cent** of all insects are known to be **phytophagous** (feeding on **plant sap and other parts of plants**). The problem is **particularly severe for plants** because, **unlike animals**, they **cannot run away** from their predators. Plants therefore have evolved an **astonishing variety of morphological and chemical defences** against herbivores.

| Defence type                 | How it works                                                                                                                                                                   | NCERT example                                                                                                                                                                                    |
|------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Morphological</b>         | <b>Thorns</b> are the <b>most common</b> morphological means of defence                                                                                                        | <b>Acacia, Cactus</b>                                                                                                                                                                            |
| <b>Chemical</b>              | <b>Many</b> plants produce and store chemicals that make the herbivore <b>sick when they are eaten, inhibit feeding or digestion, disrupt its reproduction or even kill it</b> | The weed <b>Calotropis</b> growing in abandoned fields produces <b>highly poisonous cardiac glycosides</b> - which is why you <b>never</b> see any <b>cattle or goats</b> browsing on this plant |
| <b>Chemical (commercial)</b> | A wide variety of chemical substances extracted from plants on a <b>commercial scale</b> are <b>actually</b> produced as defences against <b>grazers and browsers</b>          | <b>Nicotine, caffeine, quinine, strychnine, opium</b> , etc.                                                                                                                                     |

When **Darwin** spoke of the **struggle for existence and survival of the fittest** in nature, he was convinced that **interspecific competition is a potent force in organic evolution**. It is **generally believed** that competition occurs when **closely related species** compete for the **same resources that are limiting**, but **this is not entirely true**:

- **Firstly, totally unrelated species could also compete** for the same resource. For instance, in **some shallow South American lakes**, **visiting flamingoes** and **resident fishes** compete for their common food, the **zooplankton** in the lake.
- **Secondly, resources need not be limiting** for competition to occur; in **interference competition**, the **feeding efficiency** of one species might be **reduced** due to the **interfering and inhibitory presence** of the other species, **even if resources (food and space) are abundant**.
- Therefore, **competition** is best defined as a process in which the **fitness of one species** (measured in terms of its '**r**', the **intrinsic rate of increase**) is **significantly lower in the presence of another species**.

It is **relatively easy to demonstrate in laboratory experiments**, as **Gause** and other experimental ecologists did, that when resources are limited the **competitively superior species will eventually eliminate the other species** - but evidence for such **competitive exclusion** occurring in nature is **not always conclusive**. **Strong and persuasive circumstantial evidence does exist however in some cases**:

- **The Abingdon tortoise in Galapagos Islands** became **extinct within a decade** after **goats** were introduced on the island, **apparently** due to the **greater browsing efficiency of the goats**.
- '**Competitive release**' - another evidence for the occurrence of competition in nature. A species whose distribution is **restricted to a small geographical area** because of the presence of a **competitively superior species** is found to **expand its distributional range dramatically** when the competing species is **experimentally removed**.
  - **Connell's** elegant field experiments showed that on the **rocky sea coasts of Scotland**, the **larger and competitively superior barnacle *Balanus*** dominates the **intertidal area**, and **excludes the smaller barnacle *Chthamalus*** from that zone.
- **In general, herbivores and plants appear to be more adversely affected by competition than carnivores**.
- **Gause's 'Competitive Exclusion Principle'** states that **two closely related species competing for the same resources cannot co-exist indefinitely** and the **competitively inferior one will be eliminated eventually**. This may be true if resources are limiting, but not otherwise.

**More recent studies do not support such gross generalisations** about competition. While they **do not rule out** the occurrence of interspecific competition in nature, they point out that species facing competition **might evolve mechanisms that promote co-existence rather than exclusion**.

- One such mechanism is '**resource partitioning**'. If two species compete for the same resource, they could **avoid competition** by choosing, for instance, **different times for feeding** or **different foraging patterns**.
- **MacArthur** showed that **five closely related species of warblers** living on the **same tree** were able to **avoid competition and co-exist** due to **behavioural differences in their foraging activities**.

Considering that the **parasitic mode of life ensures free lodging and meals**, it is not surprising that parasitism has evolved in **so many taxonomic groups from plants to higher vertebrates**.

- **Many parasites have evolved to be host-specific** (they can parasitise **only a single species of host**) in such a way that **both host and the parasite tend to co-evolve**: if the host evolves special mechanisms for **rejecting or resisting** the parasite, the parasite has to evolve mechanisms to **counteract and neutralise** them, in order to be successful with the **same host species**.
- In accordance with their life styles, parasites evolved **special adaptations**: **loss of unnecessary sense organs**, presence of **adhesive organs or suckers** to cling on to the host, **loss of digestive system** and **high reproductive capacity**.
- The life cycles of parasites are **often complex**, involving **one or two intermediate hosts or vectors** to facilitate parasitisation of its **primary host**.

- The **human liver fluke** (a **trematode** parasite) depends on **two intermediate hosts (a snail and a fish)** to complete its life cycle.
- The **malarial parasite** needs a **vector (mosquito)** to spread to other hosts.

① **[NOTE]** Most parasites harm their hosts: they may reduce host survival, growth and reproduction and lower population density; they might also make the host more vulnerable to predation by weakening it physically.

| Type                    | Definition                                                                                                                                                            | Examples / features                                                                                                                                                                                                                                                                                                                                                 |
|-------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Ectoparasites</b>    | Parasites that feed on the <b>external surface</b> of the host organism                                                                                               | The <b>most familiar examples: lice on humans and ticks on dogs; many marine fish</b> are infested with <b>ectoparasitic copepods</b> ; <i>Cuscuta</i> , a parasitic plant <b>commonly found growing on hedge plants</b> , has <b>lost its chlorophyll and leaves</b> in the course of evolution and derives its nutrition from the host plant which it parasitises |
| <b>Endoparasites</b>    | In contrast, those that live <b>inside the host body</b> at different sites ( <b>liver, kidney, lungs, red blood cells</b> , etc.)                                    | Life cycles are <b>more complex</b> because of their <b>extreme specialisation</b> ; their <b>morphological and anatomical features are greatly simplified</b> while <b>emphasising their reproductive potential</b>                                                                                                                                                |
| <b>Brood parasitism</b> | A fascinating example of parasitism in <b>birds</b> , in which the <b>parasitic bird lays its eggs in the nest of its host</b> and <b>lets the host incubate them</b> | During the course of evolution, the <b>eggs of the parasitic bird have evolved to resemble the host's egg in size and colour</b> , to <b>reduce the chances of the host bird detecting the foreign eggs and ejecting them from the nest</b>                                                                                                                         |

① **[NOTE]** In brood parasitism, the cuckoo (koel) lays its eggs in the crow's nest; this occurs during the breeding season, from spring to summer.

#### (iv) Commensalism

- **Commensalism** is the interaction in which **one species benefits and the other is neither harmed nor benefited**.

| Example                                                                                                                                                                           | Who benefits                                                                                                                                                                                                                                 | Who is unaffected                                                                  |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------|
| An <b>orchid</b> growing as an <b>epiphyte</b> on a <b>mango branch</b>                                                                                                           | The orchid                                                                                                                                                                                                                                   | Neither the <b>mango tree</b> derives any apparent benefit                         |
| <b>Barnacles</b> growing on the <b>back of a whale</b>                                                                                                                            | The barnacles                                                                                                                                                                                                                                | Nor the <b>whale</b> derives any apparent benefit                                  |
| The <b>cattle egret</b> and <b>grazing cattle</b> in close association - a <b>classic example</b> , a sight you are most likely to catch if you live in <b>farmed rural areas</b> | The <b>egrets always forage close to where the cattle are grazing</b> , because the cattle, as they move, <b>stir up and flush out insects</b> from the vegetation that <b>otherwise might be difficult for the egrets to find and catch</b> | The grazing cattle                                                                 |
| <b>Sea anemone</b> that has <b>stinging tentacles</b> and the <b>clown fish</b> that lives among them                                                                             | The fish <b>gets protection from predators</b> which <b>stay away from the stinging tentacles</b>                                                                                                                                            | The <b>anemone does not appear to derive any benefit</b> by hosting the clown fish |

#### (v) Mutualism

- **Mutualism**: this interaction **confers benefits on both** the interacting species.
- **Lichens** represent an **intimate mutualistic relationship** between a **fungus** and **photosynthesising algae or cyanobacteria**.
- **Mycorrhizae** are associations between **fungi** and the **roots of higher plants**: the fungi help the plant in the **absorption of essential nutrients from the soil**, while the plant in turn provides the fungi with **energy-yielding carbohydrates**.

The most spectacular and evolutionarily fascinating examples of mutualism are found in plant-animal relationships. Plants need the help of animals for **pollinating their flowers** and **dispersing their seeds**. Animals obviously have to be paid '**fees**' for the services that plants expect from them.

- Plants offer **rewards or fees** in the form of **pollen and nectar for pollinators** and **juicy and nutritious fruits for seed dispersers**.

- Mutualisms must guard against cheaters, such as nectar thieves that do not pollinate.

- This is why plant-animal interactions **often involve co-evolution of the mutualists**, that is, the **evolutions of the flower and its pollinator species are tightly linked with one another**.

In **many species of fig trees**, there is a **tight one-to-one relationship** with the **pollinator species of wasp** (Figure 11.4). It means that a given fig species **can be pollinated only by its 'partner' wasp species and no other species**. The **female wasp** uses the fruit **not only as an oviposition (egg-laying) site** but uses the **developing seeds within the fruit for nourishing its larvae**.

1

The **wasp pollinates the fig inflorescence** while searching for **suitable egg-laying sites**.

2

In return for the favour of pollination, the **fig offers the wasp some of its developing seeds**, as **food for the developing wasp larvae**.



Fig. 11.4 (a) - Mutual relationship between fig tree and wasp:  
Fig flower is pollinated by wasp.



Fig. 11.4 (b) - Mutual relationship between fig tree and wasp:  
Wasp laying eggs in a fig fruit.

**Orchids** show a **bewildering diversity of floral patterns**, many of which have evolved to **attract the right pollinator insect (bees and bumblebees)** and **ensure guaranteed pollination by it** (Figure 11.5). **Not all orchids offer rewards**.

- The **Mediterranean orchid *Ophrys*** employs '**sexual deceit**' to get pollination done by a **species of bee**. **One petal** of its flower bears an **uncanny resemblance to the female of the bee in size, colour and markings**.

1

The **male bee is attracted to what it perceives as a female** and '**pseudocopulates**' with the flower.

2

During that process it is **dusted with pollen from the flower**.

3

When this **same bee 'pseudocopulates' with another flower**, it **transfers pollen** to it and thus **pollinates the flower**.



Fig. 11.5 - Showing bee-a pollinator on orchid flower. The original is a colour photograph; after monochrome conversion the bee still reads clearly against the pale flower and the dark background.

- ① **[NOTE]** Co-evolution in action: if the **female bee's colour patterns change even slightly** for any reason during evolution, **pollination success will be reduced unless the orchid flower co-evolves** to maintain the resemblance of its petal to the **female bee**.

- **Ecology** is the study of the relationships of living organisms with the **abiotic (physico-chemical factors)** and **biotic components (other species)** of their environment; it is concerned with **four levels** of biological organisation - **organisms, populations, communities and biomes**.
- **Evolutionary changes through natural selection take place at the population level**, hence **population ecology** is an important area of ecology.
- A **population** is a group of individuals of a given species **sharing or competing for similar resources in a defined geographical area**.
- Populations have attributes that individual organisms do not - **birth rates and death rates, sex ratio and age distribution**. The proportion of different age groups of males and females is **often presented graphically as an age pyramid**; its **shape** indicates whether a population is **stationary, growing or declining**.
- Ecological effects of any factors on a population are **generally** reflected in its **size (population density)**, which may be expressed in different ways (**numbers, biomass, per cent cover**, etc.) depending on the species.
- Populations **grow** through **births and immigration** and **decline** through **deaths and emigration**. When resources are unlimited the growth is **usually exponential**; when resources become **progressively limiting** the growth pattern turns **logistic**. **In either case, growth is ultimately limited by the carrying capacity of the environment**.
- The **intrinsic rate of natural increase (r)** is a measure of the **inherent potential of a population to grow**.
- Populations of different species in a habitat **do not live in isolation but interact in many ways**. Depending on the outcome, interactions between two species are classified as **competition** (both species suffer), **predation** and **parasitism** (one benefits and the other suffers), **commensalism** (one benefits and the other is unaffected), **amensalism** (one is harmed, other unaffected) and **mutualism** (both species benefit).
- **Predation** is a very important process through which **trophic energy transfer** is facilitated, and **some** predators help in **controlling their prey populations**. Plants have evolved **diverse morphological and chemical defenses against herbivory**.
- In competition it is presumed that the **superior competitor eliminates the inferior one** (the **Competitive Exclusion Principle**), but **many** closely related species have evolved various mechanisms which **facilitate their co-existence**. Some of the most fascinating cases of **mutualism** in nature are seen in **plant-pollinator interactions**.

NCERT gives no single-line definition, but says interacting plants, animals and microbes form communities. Thus, a **community** is **populations of different species living and interacting in one habitat** - the level above population.

The chapter supplies the equation and the constant but never demonstrates rearranging it. Using only chapter content -  $N_t = N_0 e^{rt}$  with  $e = 2.71828$ :

1

The population **doubles**, so  $N_t/N_0 = 2$ , and  $t = 3$  years.

2

Substituting:  $2 = e^{3r}$ .

3

Taking natural logarithms of both sides:  $3r = \text{natural log of } 2 = 0.6931$ .

4

So  $r = 0.6931/3 = 0.2310$  per year (about 0.231).

- ① **[NOTE]** Compare this  $r = 0.231$  with the chapter's own  $r$  values - **0.015** for the Norway rat, **0.12** for the flour beetle and **0.0205** for the human population in India in **1981**. A doubling in 3 years is a very high intrinsic rate of natural increase.